

Lecture 1: Introduction

Assessment

4 Coursework – 40%

- Mid-term Test 20%
- Quiz 5%
- Assignment (4 students/group) 15%

4 Final Examination – 60%

- Final Examination

Note: You have to get at least 40/100 in Coursework, at least 40/100 in Final Examination and at least 50/100 overall marks in order to pass this course.

Reference Books

4 Main Text:

- William Stallings (2017) Operating Systems: Internals and Design Principles. (9th ed.). Prentice Hall.
 - Digital copies are available in library!
<https://login.libezp2.utar.edu.my/login?url=https://ebookcentral.proquest.com/lib/utar-ebooks/detail.action?docID=5186478>
- Abraham Silberschatz, Peter Baer Galvin, and Greg Gagne. (2018) Operating System Concepts (10th ed.). Wiley Publishing.

4 Additional Text:

- Andrew S. Tanenbaum and Herbert Bos. (2014) Modern Operating Systems (4th ed.). Prentice Hall.
- Remzi H Arpaci-Dusseau and Andrea C Arpaci-Dusseau. (2018) Operating Systems: Three Easy Pieces (1st ed.) CreateSpace Independent Publishing Platform.

UCCD 2103 Operating Systems

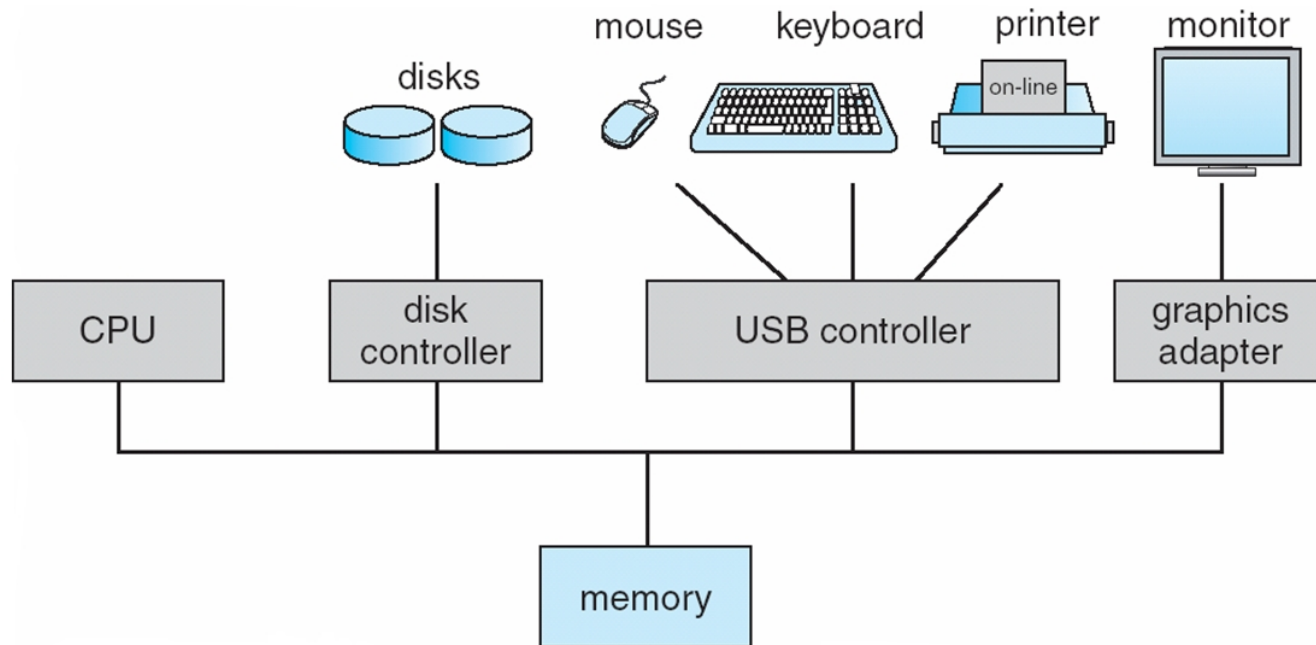
Pre-requisite? No official pre-requisite but the content will make no sense if the student does not know how to program (i.e. at least had passed Programming Concepts & Practices)

Overview

- 4 Overview of Computer-System Organization & Architecture
- 4 Computer System Structure
- 4 Computer System Operation
- 4 What is an Operating Systems?
- 4 The Role of Operating Systems
- 4 Kernel
- 4 Evolution of Operating Systems
- 4 Process Management
- 4 Memory Management
- 4 Storage Management
- 4 Protection and Security
- 4 Scheduling and Resource Management
- 4 Different Architectural Approaches
- 4 Memory Hierarchy
- 4 Summary

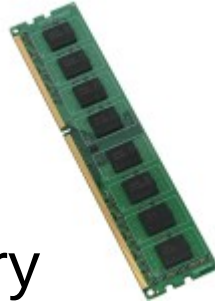
Overview of COA

- 4 One or more CPUs, device controllers connect through common bus providing access to shared memory
- 4 Concurrent execution of CPUs and devices competing for memory cycles



Overview of COA

Main
Memory



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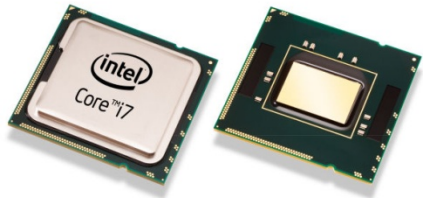
But putting together these hardware does not mean it will work automatically! You need software to manage them....

+



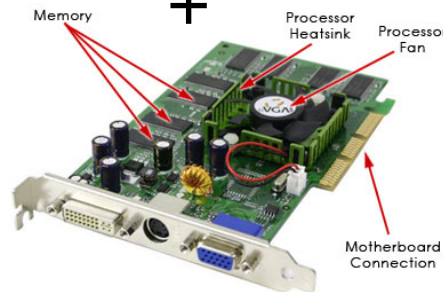
= Computer hardware

Processor



Motherboard (bus and I/O)

+

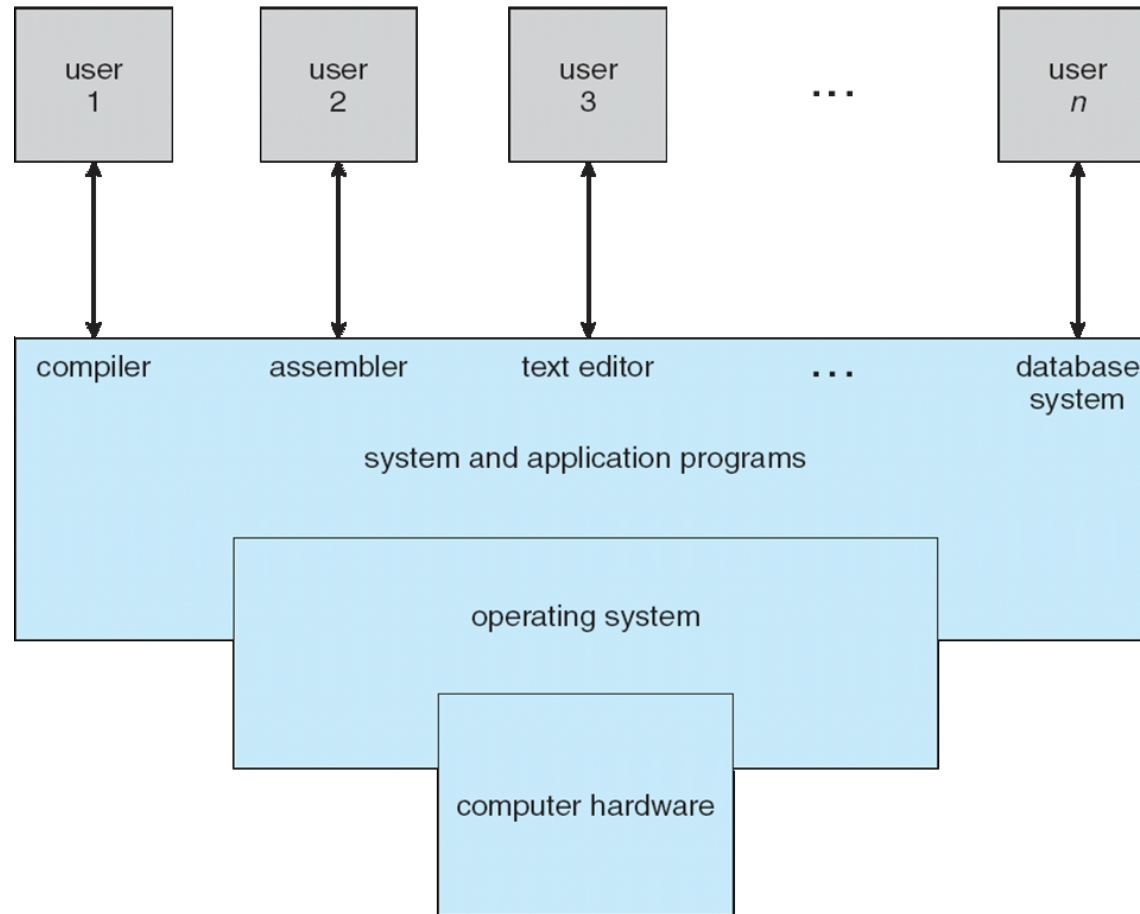


Graphics
adapter

Overview of Computer System Structure

- n Computer system can be divided into four components:
 - | Hardware – provides basic computing resources
 - 4 CPU, memory, I/O devices
 - | Operating system
 - 4 Controls and coordinates use of hardware among various applications and users
 - | Application programs – define the ways in which the system resources are used to solve the computing problems of the users
 - 4 Word processors, compilers, web browsers, database systems, video games
 - | Users
 - 4 People, machines, other computers

Overview of Computer System Structure

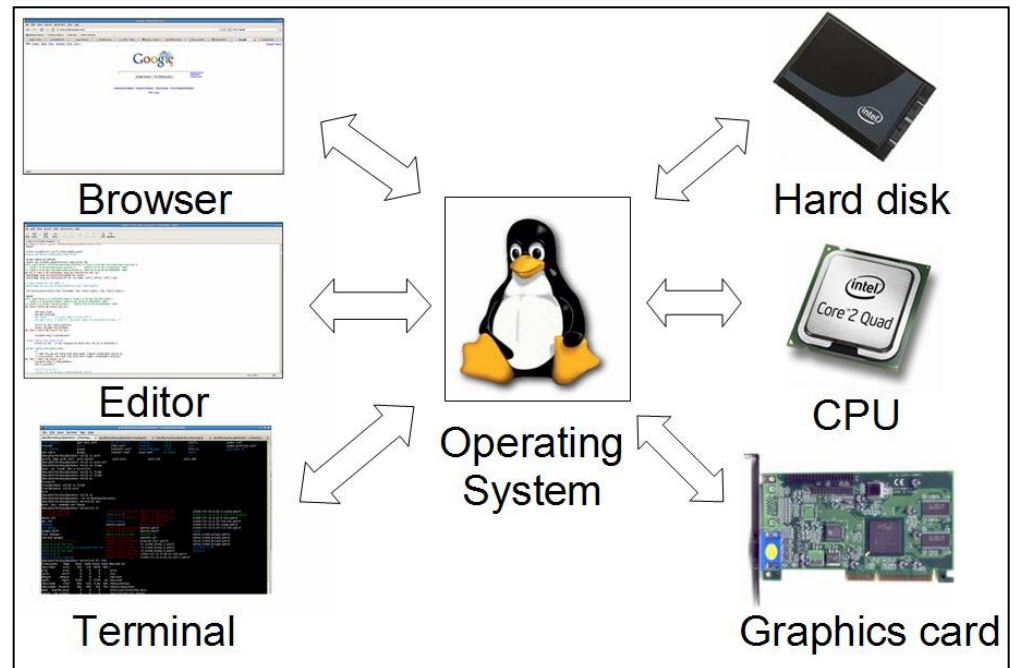


Computer-System Operation

- 4 I/O devices and the CPU can execute concurrently
- 4 Each device controller is in charge of a particular device type
- 4 Each device controller has a local buffer
- 4 CPU moves data from/to main memory to/from local buffers
- 4 I/O is from the device to local buffer of controller
- 4 Device controller informs CPU that it has finished its operation by causing an **interrupt**

What is an OS?

- 4 A program that controls the execution of application programs
- 4 An interface between applications and hardware
- 4 Main objectives of an OS:
 - Convenience
 - Efficiency
 - Ability to evolve



The Role of an OS

4 OS is a **resource manager**

- Manages all resources
- Decides between conflicting requests for efficient and fair resource use

4 OS is a **control program**

- Controls execution of programs to prevent errors and improper use of the computer
- Protect the system from incorrect or malicious programs and users

OS as Resource Manager

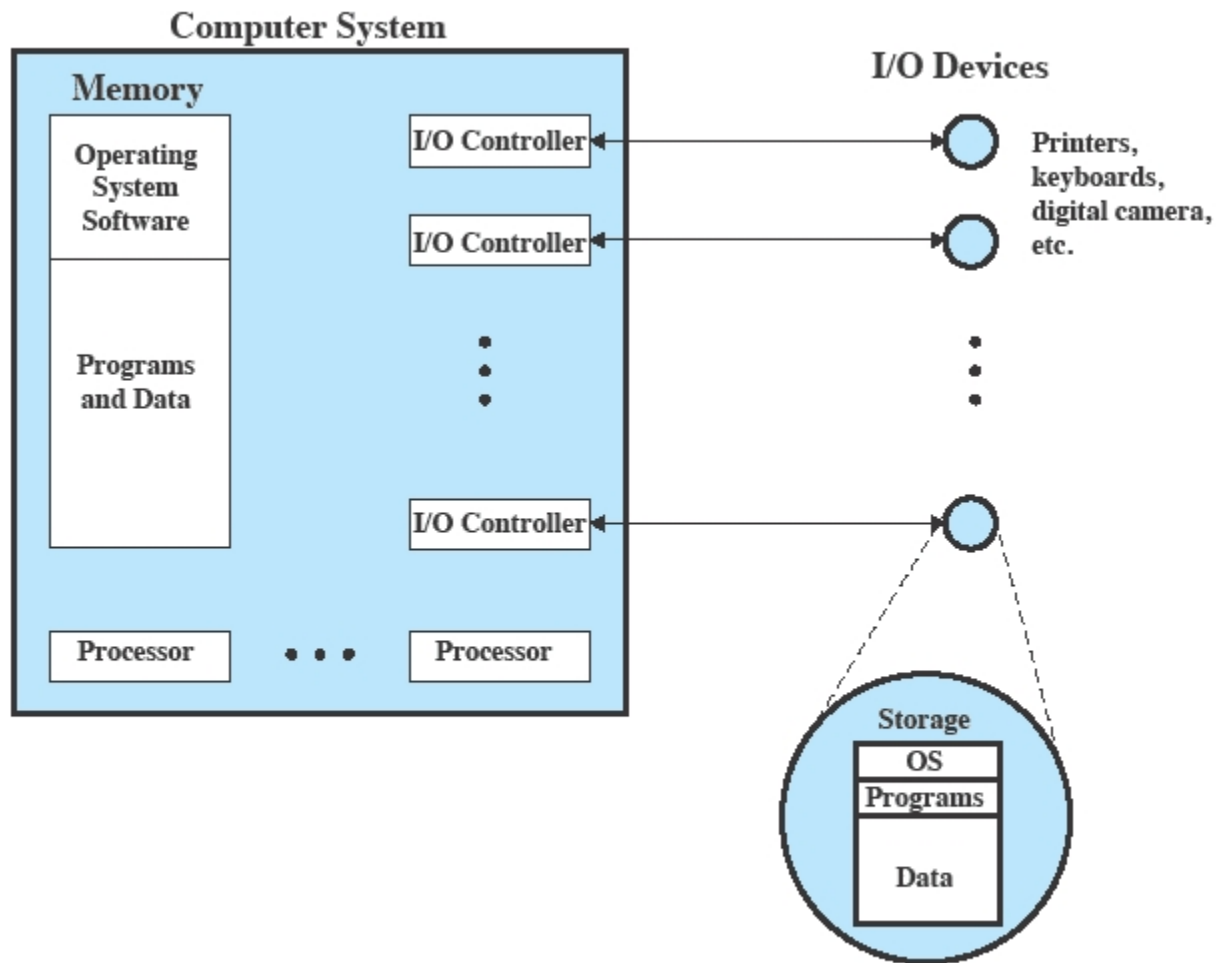


Figure 2.2 The Operating System as Resource Manager

The Role of an OS (Cont'd)

- 4 Depends on the point of view
- 4 Normal users want convenience, **ease of use**
 - Don't care about **resource utilization**
 - Shouldn't be aware whether there is other users using at the concurrently
- 4 But shared computer such as **mainframe** or **minicomputer** must keep all users happy
- 4 Users of dedicate systems such as **workstations** have dedicated resources but frequently use shared resources from **servers**
- 4 Handheld computers are resource relatively limited compared to desktop, optimized for usability and battery life
- 4 Some computers have little or no user interface, such as embedded computers in devices and automobiles (ECU)

Kernel

- 4 Fallacy: People always judge an OS from GUI point of view, which is wrong. UI is only an application that comes with the OS.
- 4 In fact, there is a lot of OS for embedded systems that does not have any GUI at all.
- 4 What we are interested in this course is the underlying software, which is called **kernel**.
- 4 “The one program running at all times on the computer” is the **kernel**. Everything else is either a system program (ships with the operating system) or an application program.

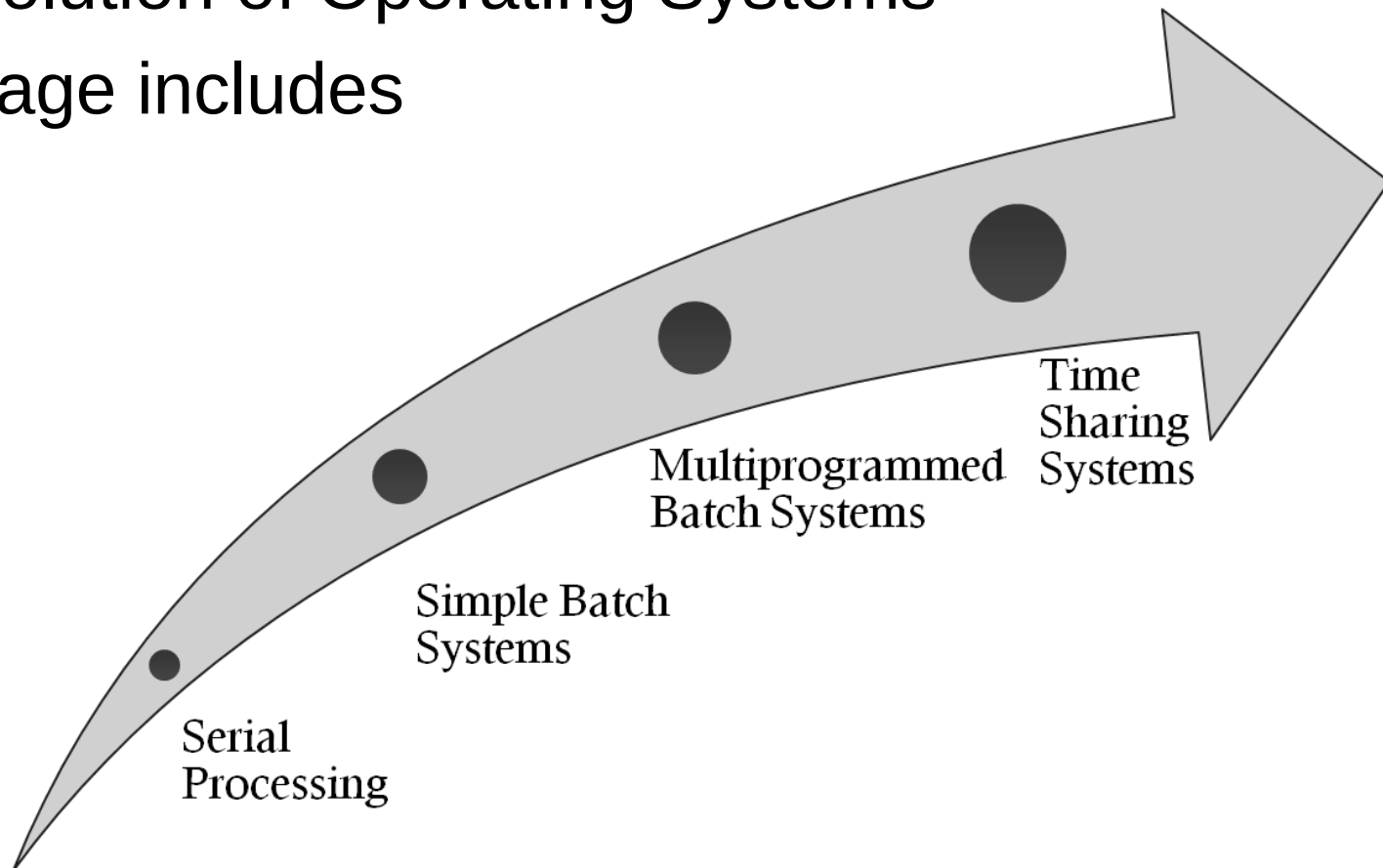
Computer Startup

- 4 **bootstrap program** is loaded at power-up or reboot
- Typically stored in ROM or EPROM, generally known as **firmware** or **BIOS**
 - Initializes all aspects of system
 - Loads operating system kernel by searching through secondary memory
 - Starts execution

Evolution of Operating Systems

4 It may be easier to understand the key requirements of an OS by considering the evolution of Operating Systems

4 Stage includes



Evolution of Operating Systems

4 Operating systems will evolve over time

- Hardware upgrades plus new types of hardware
 - e.g. support of multi-core systems through CPU scheduler, solid-state disks optimization through I/O scheduler and file-system
- New services
 - In response to user demand/the needs of system managers, the OS expands to offer new services.
 - e.g. if it is found to be difficult to maintain good performance for users with existing tools, new measurement and control tools may be added to the OS.
- Fixes
 - Any OS has faults.
 - These are discovered over the course of time and fixes are made (called **kernel patches**).
 - Of course, the fix may introduce new faults. ;-)

Serial Processing

- 4 No operating system
- 4 Machines run from a console with display lights, toggle switches, input device, and printer
- 4 Problems include:
 - Scheduling
 - Setup time

Simple batch system

4 Early computers were extremely expensive

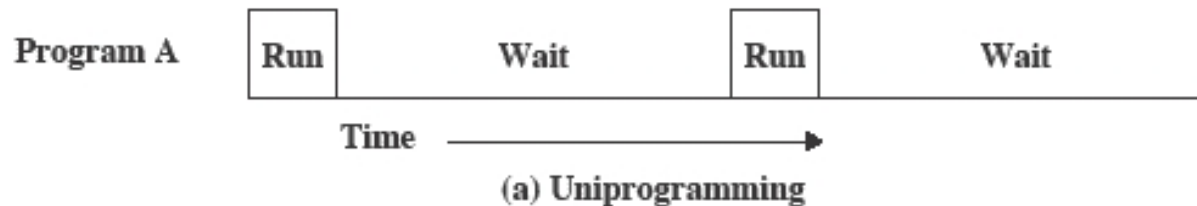
- Important to maximize processor utilization

4 Monitor

- Software that controls the sequence of events
- Batch jobs together
- Program returns control to monitor when finished

Uniprogramming

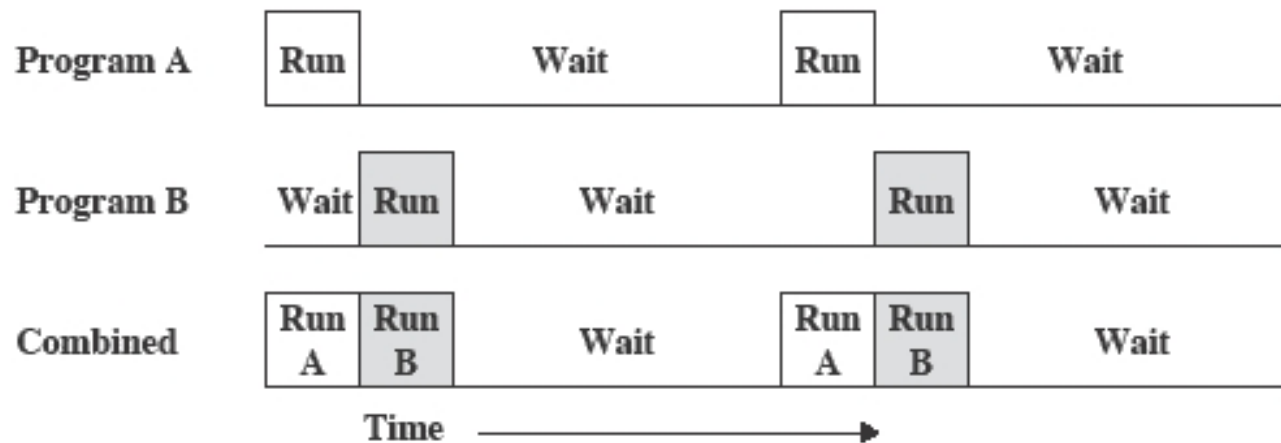
- 4 In batch system, processor must wait for I/O instruction to complete before preceding



- 4 Not efficient; the batch system can make use of the waiting time.

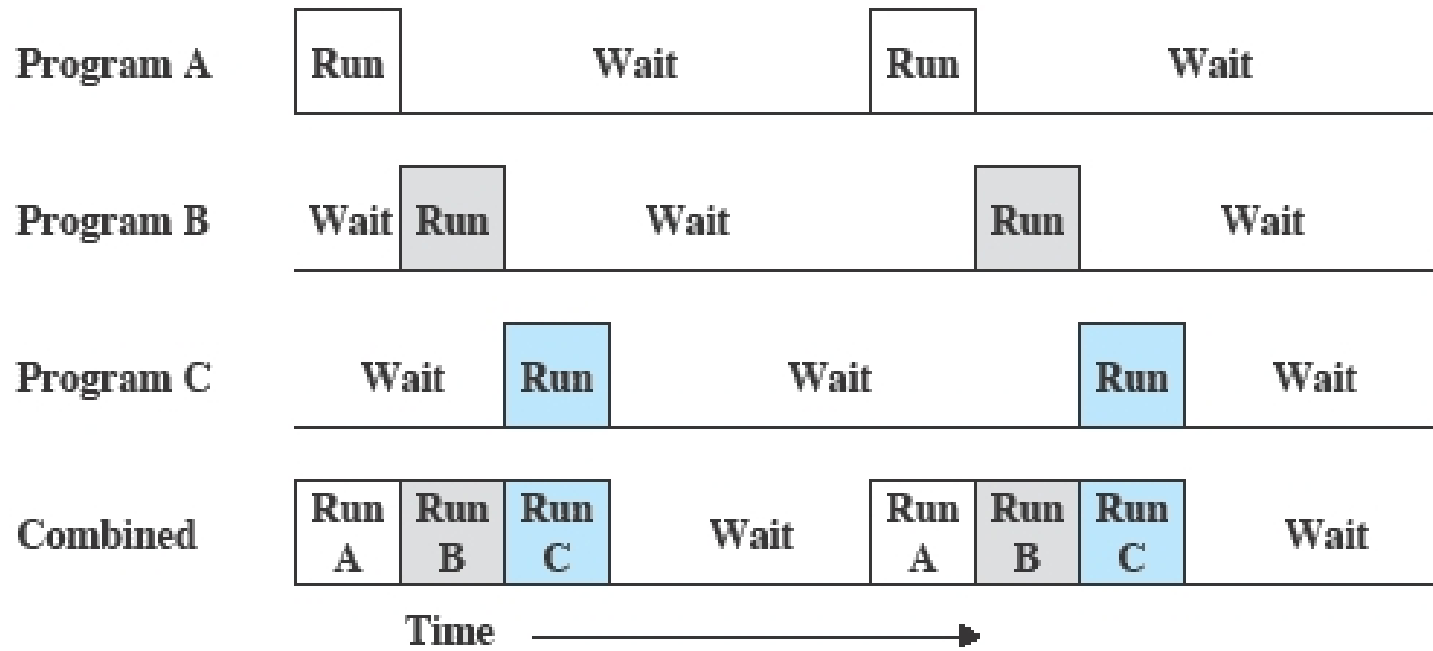
Multiprogramming

- 4 When one job needs to wait for I/O, the processor can switch to the other (ready) job



(b) Multiprogramming with two programs

Multiprogramming



(c) Multiprogramming with three programs

Time Sharing Systems

- 4 Using multiprogramming to handle multiple interactive jobs
- 4 Processor's time is shared among multiple users
- 4 Multiple users simultaneously access the system through terminals

Major Advances

4 Operating Systems are among the most complex pieces of software ever developed

4 Major advances include:

- Processes
- Memory management
- Information protection and security
- Scheduling and resource management
- System

Causes of Errors when Designing System Software

- 4 Error in designing an OS are often subtle and difficult to diagnose
- 4 Errors typically include:
 - Improper synchronization
 - Failed mutual exclusion
 - Non-determinate program operation
 - Deadlocks

Memory Management

4 The OS has 5 principal storage management responsibilities

- Process isolation
- Automatic allocation and management
- Support of modular programming
- Protection and access control
- Long-term storage

File System & Virtual Memory

4 File system implements long-term store

- Information stored in named objects, called files.
- The file is a convenient concept for the programmer and is a useful unit of access control and protection for the OS.

4 Virtual memory allows programs to address memory from a logical point of view

- Without regard to the limits of physical memory

Information Protection and Security

- 4 The problem involves controlling access to computer systems and the information stored in them.
- 4 Main issues are:
 - Availability
 - Confidentiality
 - Data integrity
 - Authenticity

Scheduling and Resource Management

- 4 Key responsibility of an OS is managing resources
- 4 Resource allocation policies must consider:
 - Fairness
 - Differential responsiveness
 - Schedule according to the requirements, e.g., schedule the process that is waiting for I/O device as soon as possible
 - Efficiency
 - Maximise throughput, minimise response time

Different Architectural Approaches

4 Various approaches have been tried, categories include:

- Microkernel architecture
- Multithreading
- Symmetric multiprocessing
- Distributed operating systems
- Object-oriented design

Microkernel Architecture

- 4 Most early OS are a monolithic kernel
 - Most OS functionality resides in the kernel.
- 4 A microkernel assigns only a few essential functions to the kernel
 - Address spaces
 - Interprocess communication (IPC)
 - Basic scheduling

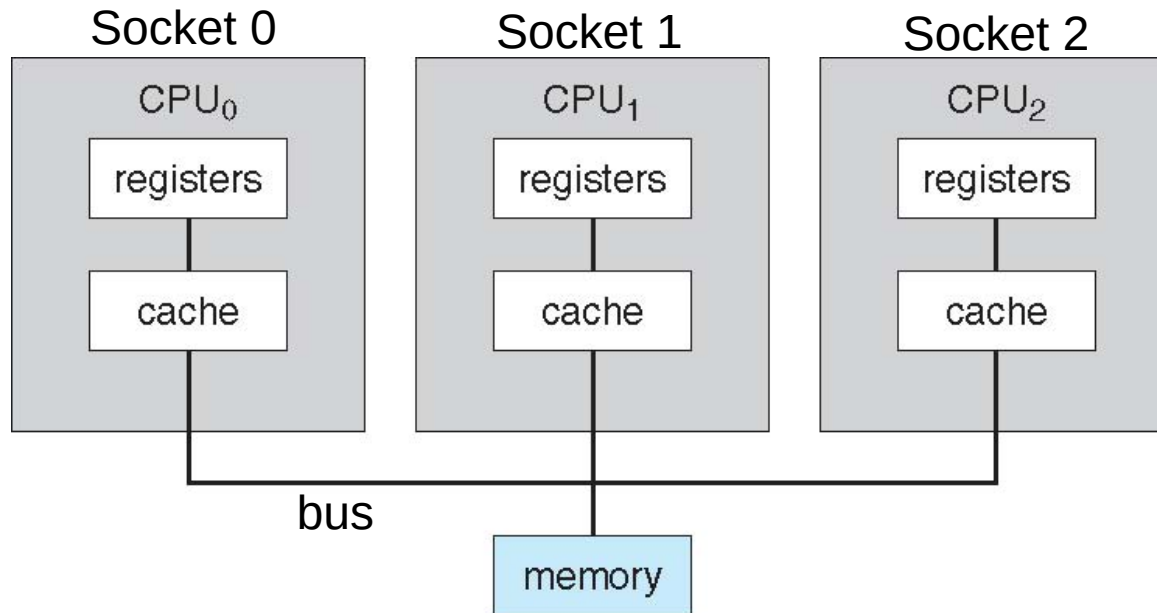
Multithreading

- 4 Process is divided into threads that can run concurrently
- 4 Thread
 - Dispatchable unit of work
 - executes sequentially and is interruptible
- 4 Process is a collection of one or more threads

Symmetric multiprocessing (SMP)

- 4 An SMP system is a stand-alone system that has:
 - Two or more similar processors of comparable capability
 - These processors share same main memory and I/O facilities
 - All processors can perform the same functions
- 4 The OS of an SMP schedules processes or threads across all of the processors.

Symmetric Multiprocessing Architecture

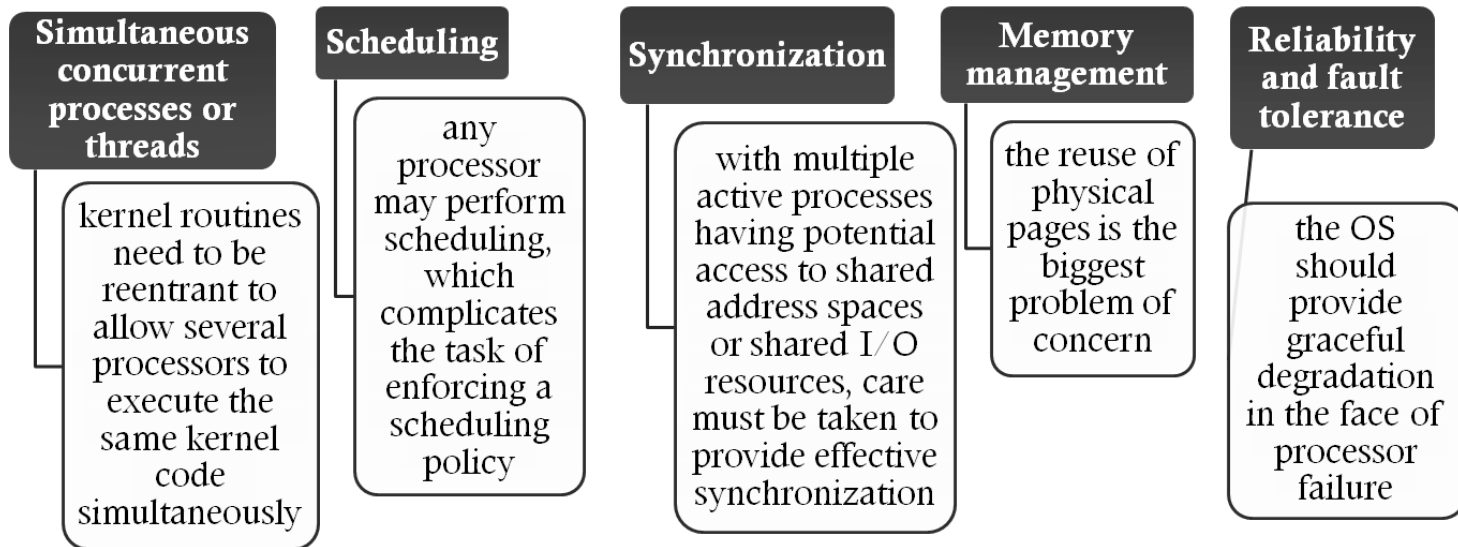


SMP Architecture

SMP OS Considerations

- 4 In an SMP system, the kernel can execute on any processor, and typically each
- processor does self-scheduling from the pool of available processes or threads.
 - The kernel can be constructed as multiple processes or multiple threads, allowing portions of the kernel to execute in parallel. The SMP approach complicates the OS.

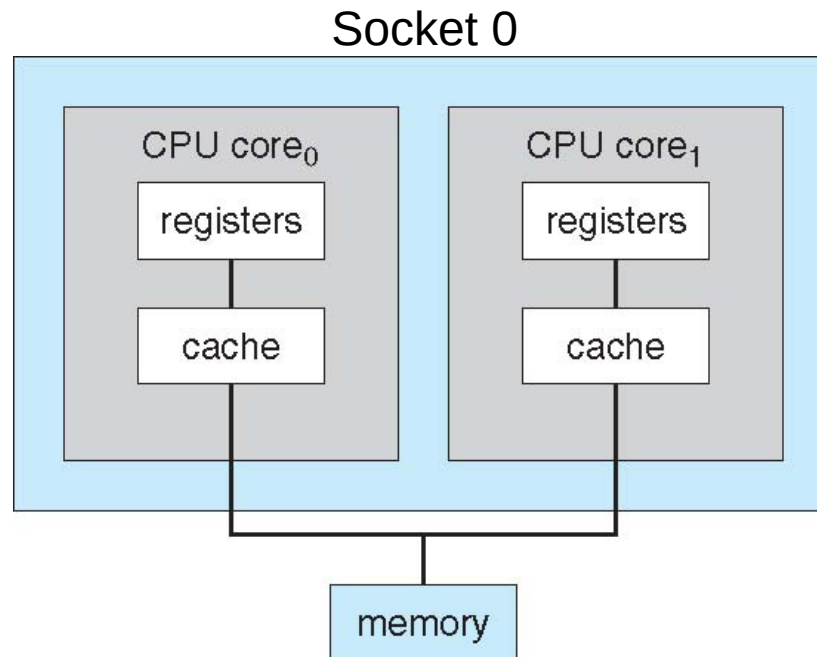
4 Key design issues:



Chip Multi-Processor Architecture

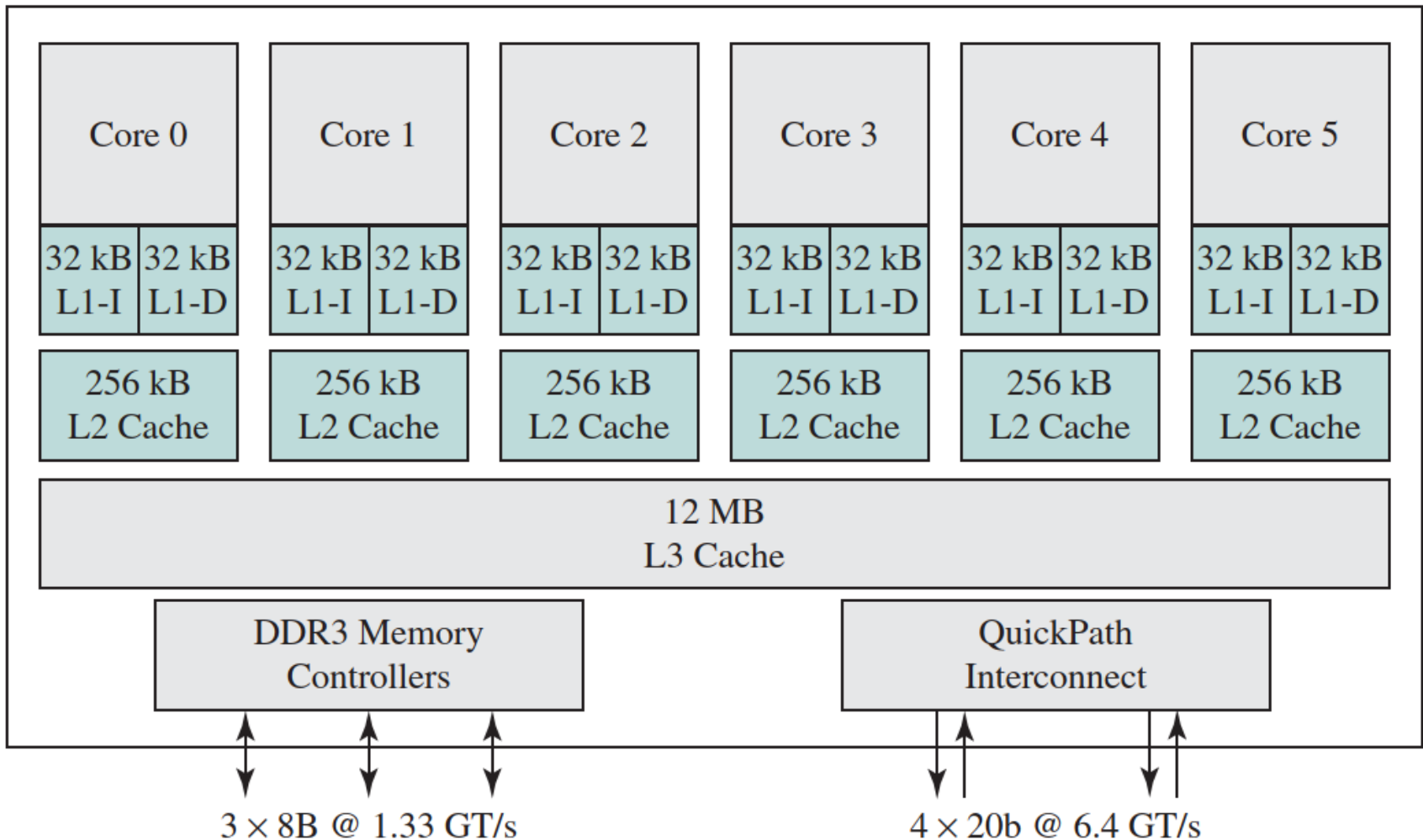
- 4 Also known as Multi-Core chip
- 4 Combines two or more processors (cores) on a single piece of silicon (die)
- 4 each core consists of all of the components of an independent processor
- 4 In addition, multi-core chips also include L2 cache and in some cases L3 cache

Chip Multi-Processor Architecture



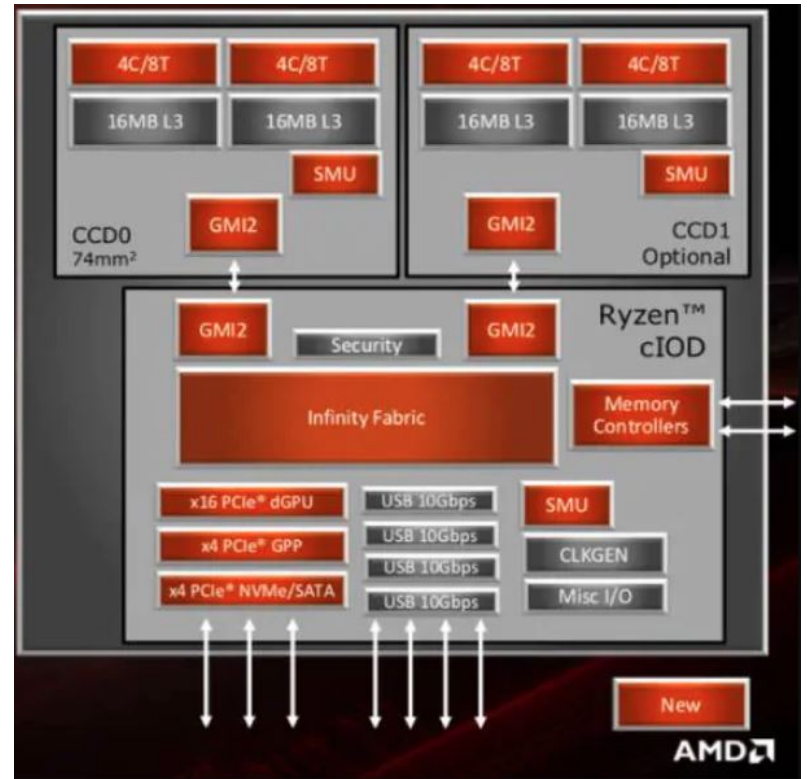
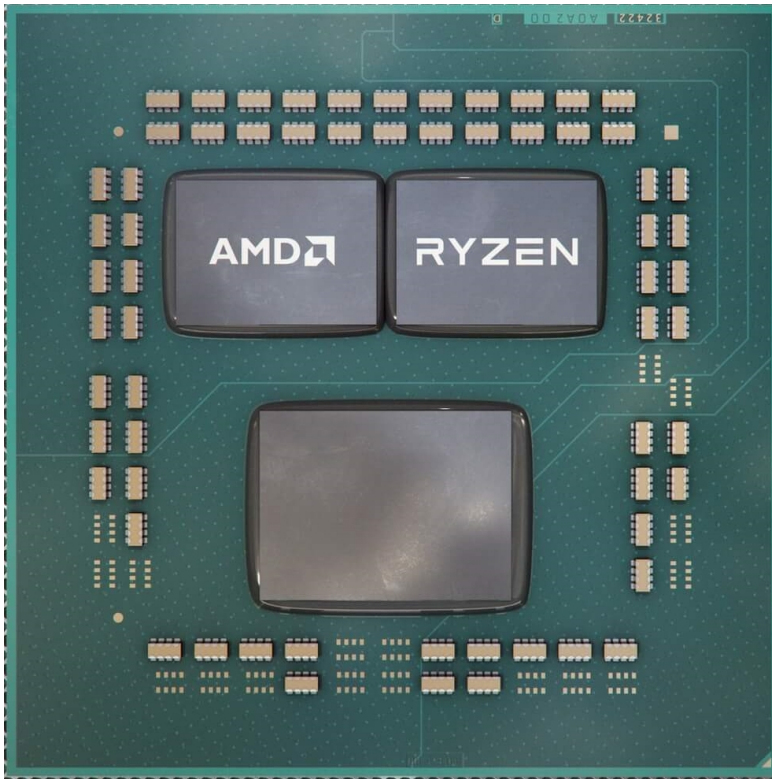
CMP: e.g. A Dual-Core Design
within a package/socket

Intel Core i7-990X



4 The L3 Cache is shared among 6 cores

AMD Ryzen 3000 series (Zen 2)



- 4 One L3 Cache is shared among 4 cores but there are four L3 caches covering a total of 16 physical cores
- 4 Quick fix to avoid performance penalty due to threads migration across different dies: https://www.legitreviews.com/game-mode-might-boost-performance-on-amd-ryzen-3900x-processors_213087

Heterogeneous multi-processor

- 4 ARM's big.LITTLE system: to which core should the OS migrate the task? Performance vs power consumption.

Fig 2: The cluster coherent model

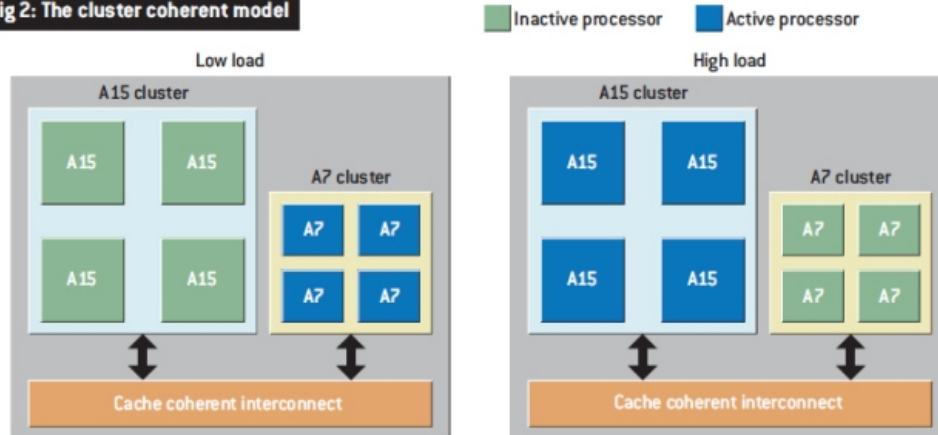
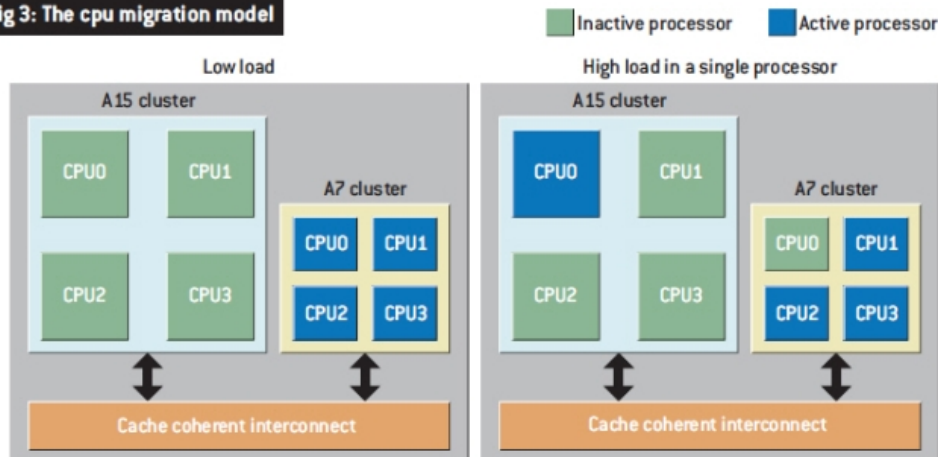


Fig 3: The cpu migration model



Multiprogramming and Multiprocessing

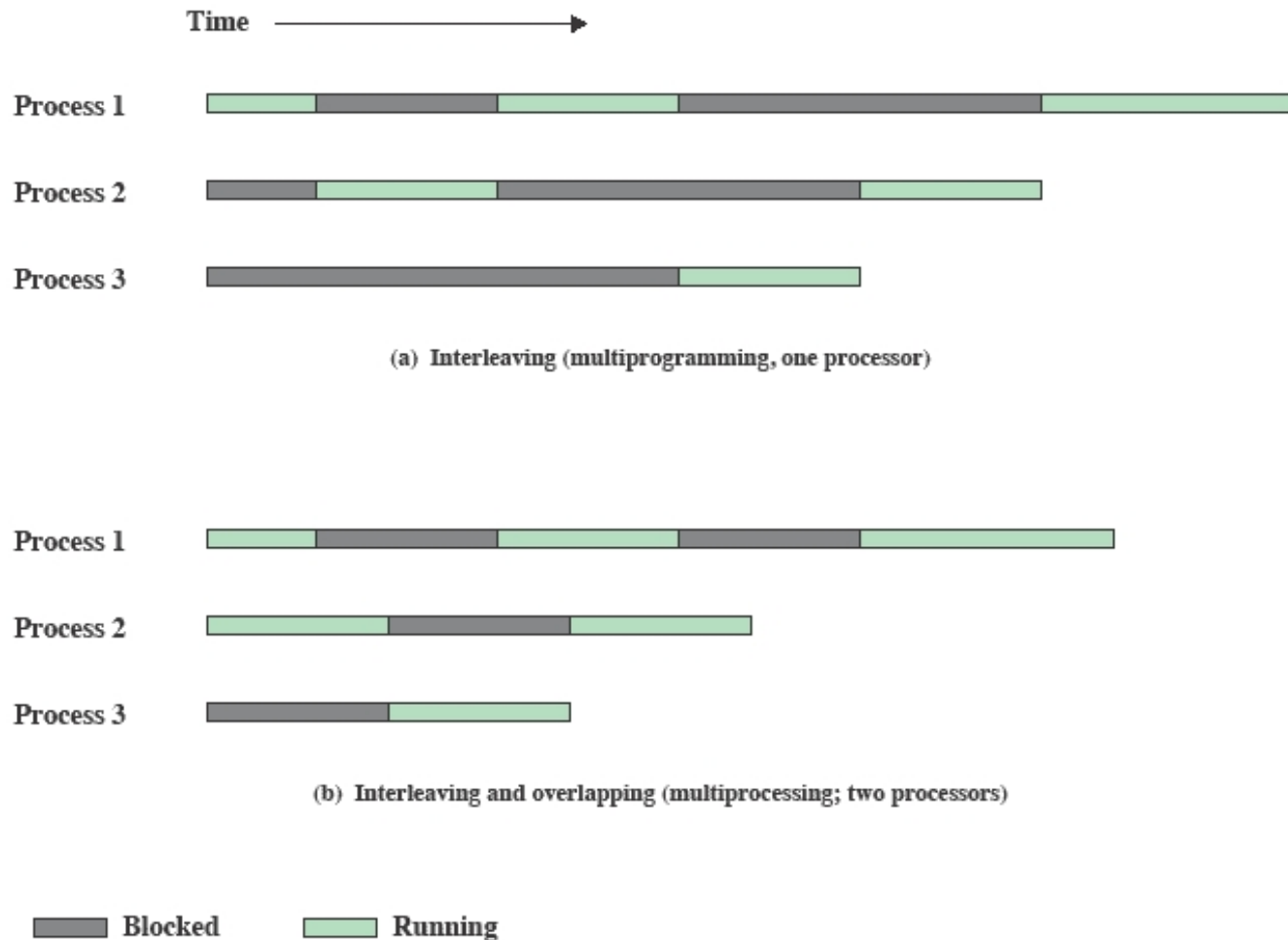
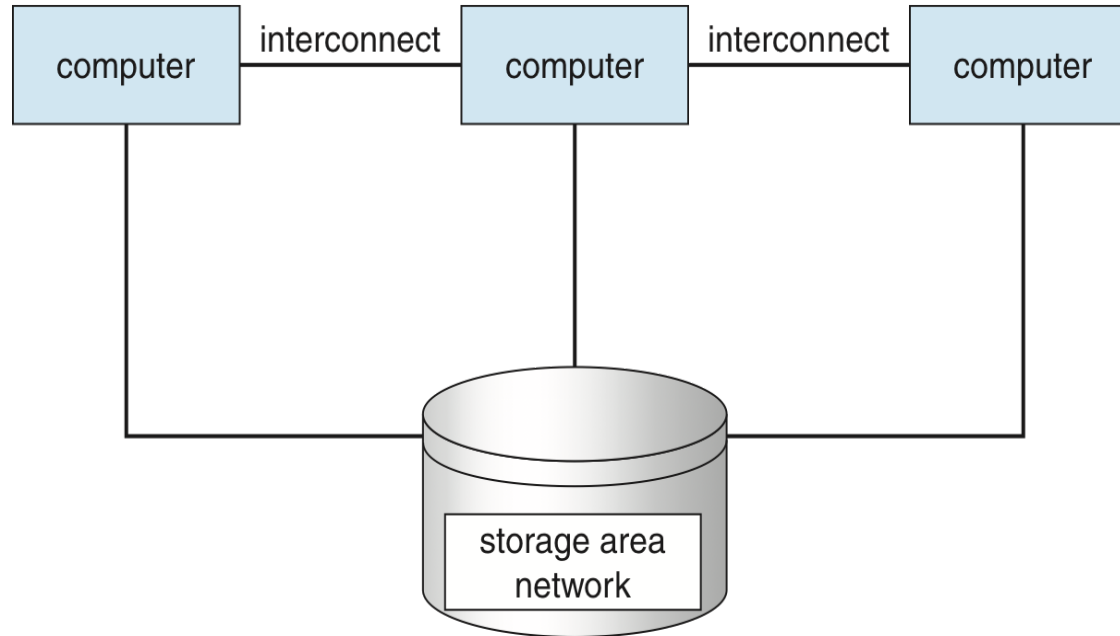


Figure 2.12 Multiprogramming and Multiprocessing

Clustered Systems



Memory Hierarchy

4 Major constraints in memory

- amount
- speed
- expense

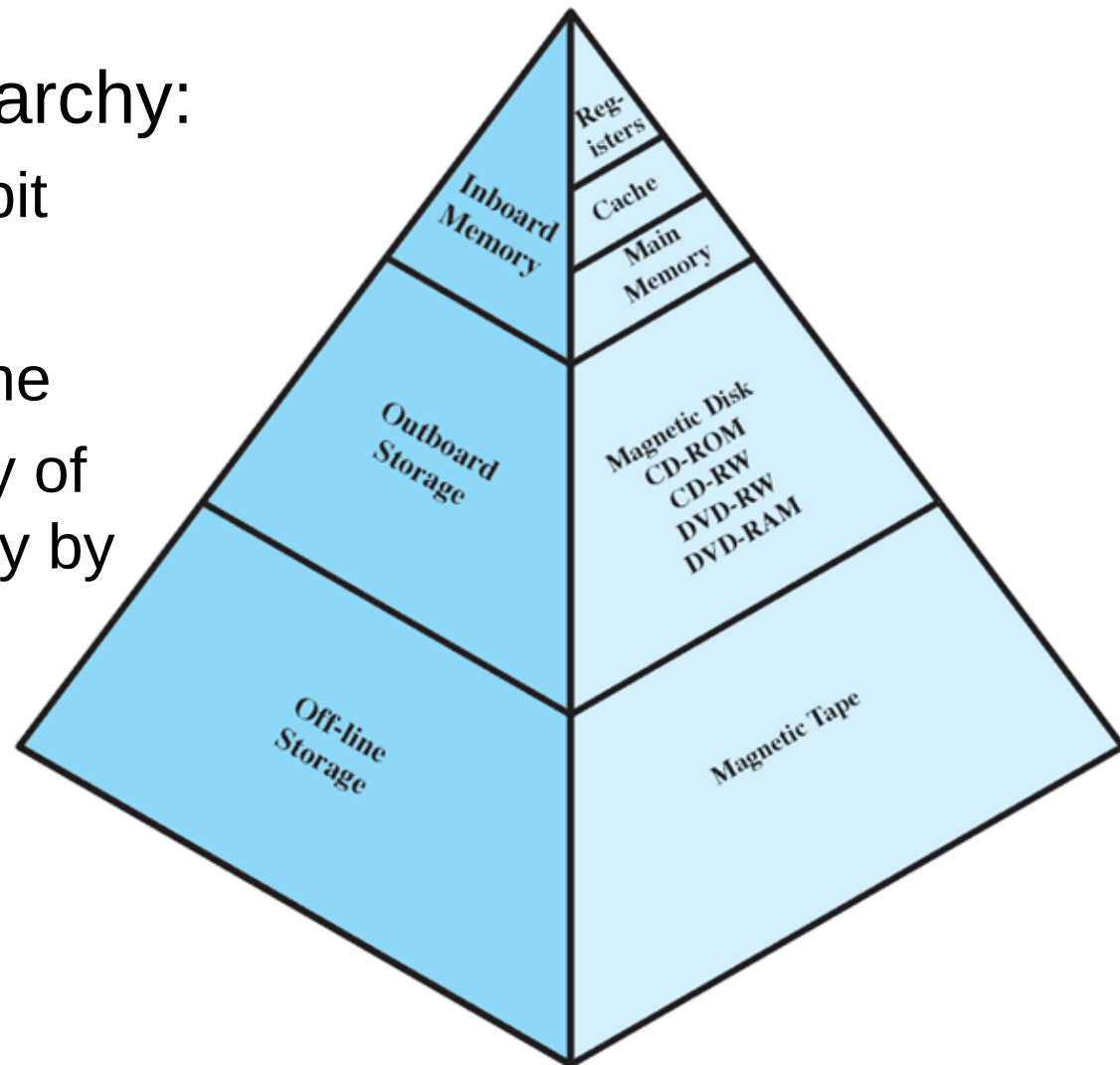
4 Memory must be able to keep up with the processor

4 Cost of memory must be reasonable in relationship to the other components

Memory Hierarchy

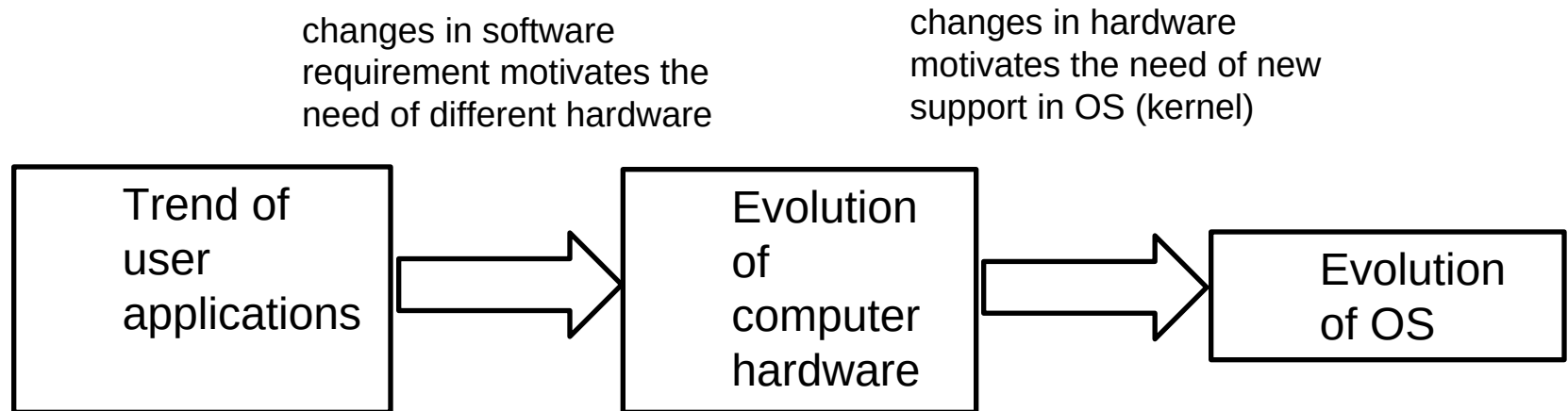
4 Going down the hierarchy:

- decreasing cost per bit
- increasing capacity
- increasing access time
- decreasing frequency of access to the memory by the processor



Summary

4 OS evolves according to:



The End